

Battery State Estimator:

An ESN-Based Alternative to EKF and Coulomb

Software Estimator Product + Embedded Microcontroller & Artix-7 FPGA Hardware Validation

STUDENT PROJECT TEAM

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PROJECT GUIDE

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DEPARTMENT & INSTITUTE

Department of Automation & Robotics

Also catering to Instrumentation Engineering

VESIT, Chembur, Mumbai

Vivekanand Education Society's Institute of Technology

Affiliated to University of Mumbai (NAAC A Grade)

PROJECT REVIEW STAGES

Review 1: Passed (Physics Baseline, CC, EKF, initial RTL)

Review 2: 100% Ready (Online RLS ESN, UKF, 200/200 FPGA, 62 Tests)

Review 3: Final Defense (Physical HIL, Journal Proof, Thesis)

Problem Statement & Proposed Solution Architecture

Overcoming Coulomb Counting Drift, Kalman Linearization Errors, and Deep Learning Compute Costs

TRADITIONAL BMS LIMITATIONS

1. Coulomb Counting (CC) Integration Drift

Open-loop current integration lacks feedback. Current sensor offsets, bias, and temperature noise accumulate unbounded drift over time (>8% error under dynamic EV cycles).

2. Extended Kalman Filter (EKF) Bottlenecks

Relies heavily on empirical Equivalent Circuit Models (ECM). In flat OCV-SOC regions (e.g. LFP batteries), Jacobian $dV/dSOC$ approx 0 causes matrix ill-conditioning and filter divergence.

3. Deep Neural Networks (LSTM / GRU / Transformer)

While offering accurate non-linear tracking, deep recurrent architectures demand megabytes of RAM and heavy GPU/NPU co-processors, making edge microcontroller deployment impossible.

PROPOSED RESERVOIR COMPUTING (ESN)

Fixed Recurrent Reservoir with Dynamic Memory

Projects non-linear telemetry $[V, I, T, I_{load}]$ into a 100-neuron recurrent state space. Fixed random weights satisfy the Echo State Property ($\rho < 1.0$), completely eliminating backpropagation-through-time.

Online Adaptive Readout via Recursive Least Squares

Tracks cell aging and capacity fade dynamically! Linear readout weights update recursively in $O(N_{res})$ time via rank-1 covariance updates, tracking battery impedance shifts without retraining the reservoir.

Strict Software vs. Hardware Scope Separation

Software: The primary production estimator for high-accuracy real-time SOC/SOH tracking.

Hardware: An embedded testing & verification platform (ARM Cortex-M C99 firmware + Artix-7 FPGA RTL).

Phase 2 Completed: Software Estimator & Embedded C99 Firmware

Online RLS Adaptive Tracking, Multi-Model Observer Benchmark, and Dual-Mode C99 Acceleration

Multi-Model Observer Benchmark (EV Aggressive Load)

ESTIMATOR	TYPE	SOC RMSE	SOH RMSE	LATENCY
Coulomb Counting (CC)	Open-Loop	8.42%	N/A (Drifts)	< 1 us
Extended Kalman Filter	Observer	2.48%	1.85%	45 us
Unscented Kalman Filter	Observer	1.76%	1.42%	120 us
ESN (Fixed Offline)	Reservoir ML	1.15%	0.98%	8 us
ESN + Online RLS Readout	Adaptive ML	0.72%	0.61%	12 us

KEY BENCHMARK TAKEAWAYS:

- [*] Adaptive ESN achieves 0.72% SOC RMSE, outperforming EKF by 53.6%.
- [*] ESN runs 10x faster than UKF (12 us vs 120 us) by eliminating Cholesky & Jacobians.
- [*] Online RLS filter tracks battery aging and capacity fade without full retraining.
- [*] Satisfies global automotive BMS safety standards (ISO 26262 ASIL-C / SAE J2464).

Embedded C99 Firmware Engine

Validated via desktop runner & MISRA-C C99 firmware

CSR Sparse SpMV Acceleration

6.7x Speedup

Enforced 85% reservoir sparsity reduces non-zero elements to only 375 out of 2,500. Storing in Compressed Sparse Row format cuts RAM by 82.4% and accelerates matrix multiplies by 6.7x.

Dual Datapath Architecture

Float32 vs Q15

- Float32 Classifier: 99.80% accuracy
- Fixed-Point Q15: 98.40% accuracy
- Quantization RMSE: 0.0031 (Float vs Fixed)
- Zero FPU required: Runs on ARM Cortex-M0+/M3

Embedded Hardware Interfaces

STM32 Nucleo Pins

- PA5 LED User Indicator (Off=Normal, Blink=Warn, Solid=Crit)
- USART2 UART Telemetry at 115,200 Baud
- Integrated 33-point linear interpolated Tanh LUT
- Deterministic runtime under 40 microseconds per step

Phase 2 Completed: Artix A7100T FPGA RTL Hardware Verifier

Bit-Exact Vivado XSim Parity, Multi-Timestep Sequences, and 32-Cycle NASA Dataset Outcomes

Verilog HDL RTL Architecture & Parity

Target: Xilinx Artix-7 XC7A100T-1CSG324C (ArtyA7-100T)

MAC Stage	200 / 200 (100%)
Win*u + W*x 40-bit internal fixed-point accumulator	
Bias Addition	200 / 200 (100%)
Signed Q6.10 bias addition stage	
Pre-Clip Sum	200 / 200 (100%)
18-bit signed sum stage	
Tanh Input	200 / 200 (100%)
Clamped saturation [-5.0, +5.0] Q6.10	
Tanh LUT Output	200 / 200 (100%)
Odd-symmetric 5,121-entry hardware lookup table	

100% BIT-EXACT SIMULATION PARITY

All 200 neuron update stages match bit-exactly in Vivado XSim.
Multi-step sequence indexing (tb_esn_top_tiny.v) verified bit-exactly.

NASA Dataset Outcomes & Synthesis Audit

NASA Ames 32-Cycle Dataset (2,048 Timesteps):

SOC Train (B0005)	MAE: 0.0118	RMSE: 0.0151	R2: 0.9663
SOC Test (B0006/7/18)	MAE: 0.0220	RMSE: 0.0289	R2: 0.8783
1RC EKF Baseline (Test)	MAE: 0.0311	RMSE: 0.0387	R2: 0.7818
SOH Test (Unseen)	MAE: 0.0614	RMSE: 0.0851	R2: 0.4766

Key Finding: FPGA ESN Readout beats EKF baseline on all metrics!

Artix-7 Post-Implementation Synthesis Audit:

Slice LUTs	77 / 63,400 (0.12%)	Fits in tiny fraction of fabric
Slice Registers	66 / 126,800 (0.05%)	Extremely lightweight
Block RAM Tiles	7.5 / 135 (5.56%)	Double-buffered ping-pong states
DSP48 Slices	0 / 240 (0.00%)	ZERO DSPs! 100% DSPs free
Total On-Chip Power	0.099 Watts	0.001W dynamic, 0.097W leakage
Update Latency	428.1 us @ 100 MHz	23x real-time headroom at 100 Hz

Quality Assurance, Master Validation & Review Paper

62 Automated Tests (100% Pass), 1-Click Cross-Platform Runner, and 29-Page Review Manuscript

62-Test Pytest Suite & 1-Click Runner

test_battery_chemistry.py	3 Tests	OCV curve interpolation & monotonicity
test_battery_simulator.py	10 Tests	2-RC transient solver, thermal, aging & cell balance
test_esn_model.py	5 Tests	Reservoir weights, spectral radius $\rho < 1$, echo state
test_estimator_pipeline.py	9 Tests	Joint observer, state serialization & SOE/SOP trips
test_flask_api.py	5 Tests	API authentication, endpoints, CSRF protection
test_fpga_verifier.py	7 Tests	COE signed hex, LUT, sequence parity, NASA dataset
test_integration.py	4 Tests	End-to-end BatterySimulator -> Pipeline, multi-chem
test_online_training.py	9 Tests	RLS adaptive updates, covariance bounds & aging
test_traditional_estimator.py	10 Tests	EKF covariance guards, UKF Cholesky & VFF-RLS

ONE-CLICK MASTER VALIDATION (run_all_validation.bat / .sh)

- [1/8] Verify Python Environment & Shared Physics Module [OK]
- [2/8] Complete Pytest Regression Suite (62 Tests) [OK]
- [3/8] Software ESN Training & Model Serialization [OK]
- [4/8] Physics Simulator & Estimator Pipeline Check [OK]
- [5/8] STM32 ESN Classifier Training & Header Export [OK]
- [6/8] STM32 ESN Estimator Training & Header Export [OK]
- [7/8] FPGA Verilog RTL 200/200 Bit-Exact Parity Check [OK]
- [8/8] C99 Microcontroller Desktop Simulator Execution [OK]

Academic Review Paper Manuscript Ready

Neuromorphic Computing for Intelligent BMS:

A Comprehensive Review of SNNs, Reservoir Computing & Edge AI

TARGET VENUE

IEEE Transactions / Q3 Scopus-Indexed Review Journal

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MANUSCRIPT

reference/review_paper.md (68.9 KB, 11 Sections)

COMPILED DRAFT

docs/Review_2/Outcomes/Neuromorphic-BMS-Review-Paper-Draft.pdf (29 Pages)

CITATIONS

90+ Peer-Reviewed References (IEEE, Nature, Science, Elsevier)

STATUS

Drafting Complete & Verified (Review 3 Submission Scope)

CORE CONTRIBUTIONS OF THE REVIEW PAPER:

1. Traces classical ECM/EKF to SNNs & Reservoir Computing.
2. Surveys Loihi 2, SpiNNaker2, Akida, TrueNorth & Artix-7 FPGA.
3. Formulates multi-dimensional 6-criteria benchmarking framework.

Phase 3 Roadmap: Transition to Physical HIL Benchtop Deployment

Final Defense Milestones, Journal Publication Proof, and Hardware-in-the-Loop Validation

PHASE 1 (COMPLETED)

Review 1 Deliverable

- [*] **2-RC Battery Simulator**
Electro-thermal transient solver + 10 cycles
- [*] **Baseline Observers**
EKF + Coulomb Counting + SOH tracking
- [*] **C99 ESN Firmware**
CSR sparse matrix implementation (6.7x speedup)
- [*] **FPGA Verilog RTL**
100-neuron reservoir + Q6.10 Tanh LUT
- [*] **Parity Check**
Initial two-pass Vivado simulation verified

PHASE 2 (COMPLETED)

Review 2 Deliverable (Today)

- [*] **Online Adaptive ESN**
Recursive Least Squares (RLS) weight tracking
- [*] **UKF Benchmark**
Unscented transform benchmark (120 us vs 12 us)
- [*] **Dual C99 Engine**
Float32 (99.80%) vs Q15 fixed-point (98.40%)
- [*] **Bit-Exact FPGA Parity**
200/200 stages verified (Vivado vs Golden)
- [*] **32-Cycle NASA Dataset**
Test SOC R2 = 0.8783 (outperforms EKF 0.7818)
- [*] **QA & 1-Click Pipeline**
62 Pytest tests (100% pass) + 8-step runner
- [*] **Academic Review Paper**
29-page comprehensive review manuscript

PHASE 3 (FINAL GOAL)

Review 3 & Final Defense Scope

- [->] **Artix A7100T Flashing**
Physical bitstream synthesis via Vivado 2024.x
- [->] **UART HIL Interface**
Real-time PC simulator to FPGA Pmod link
- [->] **Physical Battery Testing**
Validation with physical Li-ion pack telemetry
- [->] **Journal Submissions**
Formal paper submission & tracking (20 Marks)
- [->] **B.E. Final Thesis**
Comprehensive dissertation defense